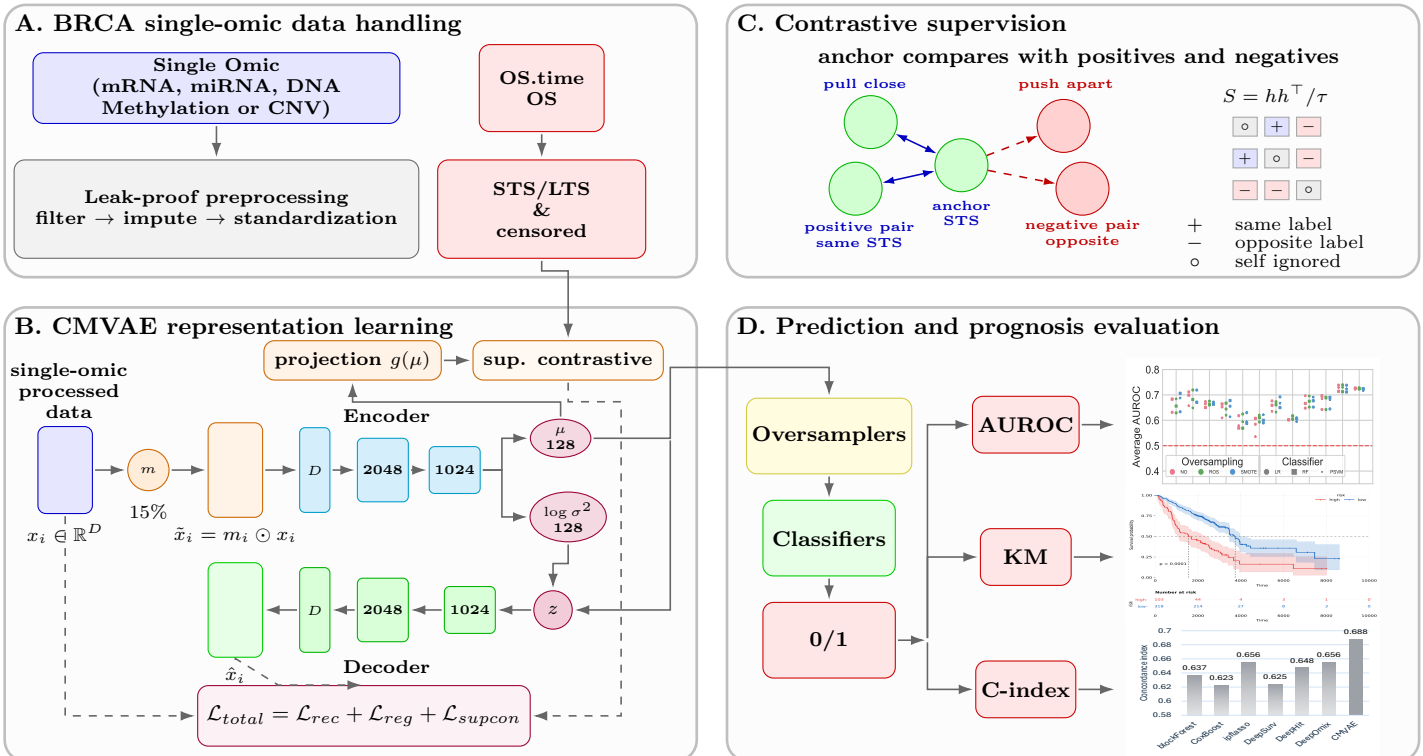

A. BRCA single-omic data handling
Single Omic (mRNA, miRNA, DNA Methylation or CNV)
OS.time OS
Leak-proof preprocessing filter → impute → standardization
STS/LTS & censored
B. CMVAE representation learning
single-omic processed data
projection $g(\mu)$
sup. contrastive
Encoder
m
15%
$x_i \in \mathbb{R}^D$
$\tilde{x}_i = m_i \odot x_i$
D
2048
1024
μ 128
$\log \sigma^2$ 128
z
$\hat{x}_i$
Decoder
$\mathcal{L}_{total} = \mathcal{L}_{rec} + \mathcal{L}_{reg} + \mathcal{L}_{supcon}$
C. Contrastive supervision
anchor compares with positives and negatives
pull close
push apart
positive pair same STS
anchor STS
negative pair opposite
$S = hh^\top/\tau$
+ same label
− opposite label
○ self ignored
D. Prediction and prognosis evaluation
Oversamplers
Classifiers
0/1
AUROC
KM
C-index
Average AUROC
Oversampling
Classifier
Concordance Index
0.637
0.623
0.656
0.625
0.648
0.656
0.688
MoCoFreat
CoxBoost
lightaso
DeepSurv
DeepFHt
DeepOmix
CMvAE